

UTSAV CHAUDHARY

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Senior **DevOps Engineer** with 7+ years of hands-on experience building, automating, and operating **production-grade AWS and Azure infrastructure** for high availability systems in regulated and enterprise environments, growing from build, release, and backend engineering into full platform ownership. Proficient in **Kubernetes and Helm orchestration, Infrastructure as Code with Terraform and Ansible, CI/CD with Jenkins, GitHub Actions, and Azure DevOps, GitOps with Argo CD, and multicloud deployments** across Dev, Test, UAT, QA, and Production. Experienced leading **disaster recovery, observability, incident response**, and compliance-ready platform operations (**HIPAA, GxP**) that meet **SLA, RTO, and RPO** targets for business-critical systems, while partnering with development teams on **Python** automation, **secure coding**, and release quality.

Technologies

Cloud & Infrastructure: AWS (EC2, EKS, ECS, Fargate, Lambda, RDS, S3, EBS, VPC, IAM, CloudWatch, API Gateway, Secrets Manager, Route 53, ALB/ELB, CodeDeploy, KMS, CloudHSM), Azure (App Services, AKS, Azure DevOps, Azure SQL, Application Gateway, Front Door, Key Vault, Entra ID), Docker, Kubernetes, Helm, Rancher, Argo CD

IaC & Automation: Terraform, Ansible, Puppet, Chef, Bash, Python, PowerShell, YAML, CloudFormation, Nginx, Cron

CI/CD & Version Control: Jenkins, GitHub Actions, Bitbucket Pipelines, AWS CodePipeline, Azure DevOps, Git, GitHub, GitLab, Bitbucket, AWS CodeArtifact

Monitoring & Logging: Prometheus, Grafana, Elasticsearch, Logstash, Kibana, CloudWatch, Application Insights, Datadog, Elastic Stack

Networking & Security: VPCs, VNets, Security Groups, NSGs, IAM Policies, DNS, VPN, LDAP, RBAC, SSL/TLS, OAuth 2.0, HIPAA, GxP, PCI, OWASP, vulnerability assessments, secrets management, key rotation

Practices: GitOps, canary releases, environment promotion, GitFlow and branching strategies, release management, on-call, incident response, root cause analysis, runbooks, SLA/SLO tracking, disaster recovery (RTO/RPO), cost optimization, code reviews, automated testing

Development & Tools: Python, TypeScript, JavaScript, Groovy, FastAPI, Node.js, React.js, Next.js, REST APIs, gRPC, WebSockets, PostgreSQL, MongoDB, Redis, Firebase, Kafka, Jira, Confluence

Operating Systems: Linux (Ubuntu, CentOS, RHEL), Windows Server, macOS, Raspberry Pi

Experience

Sr. DevOps Engineer, Hybrid, **Labs196**, TX

Feb 2025 - Present

- Collaborated with platform and compliance teams to support a **multicloud strategy** across **AWS and Azure**, helping deploy and maintain **Kubernetes** workloads on **EKS and AKS** with **Helm** across **Dev, Test, UAT, QA, and Production** for regulated lab and research platform services.
- Contributed to **GitOps** release workflows using **Argo CD**, collaborating with platform and staff engineers to standardize application deployments and reduce manual release steps across **GxP-validated** environments.
- Supported centralized cluster operations with **Rancher** managing **EKS and AKS** clusters, integrating **Azure Entra ID** for **RBAC**, least-privilege access, and audit-ready logging aligned with **FDA** and internal compliance reviews.
- Maintained multicloud infrastructure through **Terraform** and **Ansible**, managing **AWS** resources including **VPCs, security groups, IAM policies**, and **RDS**, and **Azure** resources including **VNets, NSGs, Key Vault**, and **Azure SQL** with version-controlled **IaC**.
- Contributed to cross-cloud **observability** using **Prometheus, CloudWatch Logs**, and the **ELK Stack**, supporting on-call incident response and production **SLA** targets for platform and data pipeline services.
- Helped automate backup and retention workflows for **S3, Azure Blob Storage**, and cloud-hosted databases, following **GxP** retention requirements established by compliance and platform leadership.
- Participated in **on-call** rotations and **incident response**, creating **runbooks** and coordinating with engineering teams to troubleshoot and restore production services across both cloud environments.
- Built and maintained **CI/CD pipelines** in **GitHub Actions** and **Bitbucket Pipelines** with unit tests, **Docker** image builds, image and dependency **security scanning**, **Terraform plan/apply** gates, and **environment promotion** from **Dev** through **Prod**.
- Provided technical leadership on parallel platform workstreams, leading **design reviews** and **code reviews** for pipeline, **IaC**, and **Helm** chart changes so teams stayed consistent on **branching strategies**, versioning, and documentation standards.
- Containerized and deployed **Python** and **Node.js** microservices, **AI agents**, and **MCP servers** onto **EKS/AKS**, managing **Kubernetes** secrets, config maps, **HPA** auto scaling, resource limits, and health probes for reliable rollouts.
- Managed secrets and credentials for **Stripe, HubSpot, Sumsb**, and **Azure OpenAI** integrations through **AWS Secrets**

Manager and Azure Key Vault, applying **PCI-aligned** access controls and key rotation.

- Led **root cause analysis** for production incidents using **Prometheus**, **Grafana**, and **CloudWatch** metrics and logs, publishing postmortems and follow-up automation that reduced repeat alerts.
- Wrote **Python** and **Bash** automation for environment bootstrapping, cluster maintenance, log rotation, and cost reporting, and tuned node groups and storage classes to control **AWS** and **Azure** spend.

DevOps Engineer, On-Site, **Delta Dental**, GA

Jan 2024 - Jan 2025

- Collaborated with a distributed platform team, including **California-based engineers**, to support production **Azure Kubernetes Service (AKS)** clusters running **containerized claims and payor workloads** behind **Azure Application Gateway**, helping maintain **99.9% availability** for **member eligibility** and **claims processing** services.
- Contributed to centralized **monitoring and alerting** using **Application Insights**, **Azure Monitor**, and **Grafana dashboards**, integrated with **Azure Front Door** health probes and gateway metrics, supporting a team reduction in **mean time to detect (MTTD)** from **30 minutes** to **5 minutes** across **claims adjudication** and **enrollment** workflows.
- Assisted in migrating legacy applications to **cloud-native AKS** on Azure, working with senior DevOps engineers to improve **resource utilization** by **40%** for the **member enrollment** and **provider portal** platforms.
- Supported **canary release** rollouts through **Azure DevOps** pipelines with **Application Gateway** traffic splitting under senior guidance, contributing to a **75%** reduction in deployment-related incidents for the **member self-service portal**.
- Helped automate **backup** and **disaster recovery (DR)** for **Azure SQL** and cloud-hosted data services, supporting team targets of **RPO 10 minutes** and **RTO 30 minutes** for critical **claims transaction processing** systems.
- Supported **HIPAA-aligned** security practices using **Azure Key Vault** for encryption and key rotation of **member PHI and claims data** at rest and in transit, following standards set by senior engineers and the compliance team.
- Maintained **Terraform** modules for **AKS**, **Azure SQL**, **Application Gateway**, **Front Door**, **VNets**, and **NSGs**, with remote state, pull request plan reviews, and drift detection across environments.
- Supported **environment promotion** across **Dev**, **Test**, **UAT**, and **Prod** through **Azure DevOps** multi-stage pipelines with approval gates, **Git workflows**, and **branching strategies**, keeping **Angular**, **React**, **Python**, and **Node.js** service releases consistent.
- Containerized **FastAPI/Flask** and **Node.js** claims and eligibility APIs with **Docker**, adding **automated testing** and image scanning to pipelines before **AKS** promotion.
- Led **root cause analysis** and **vulnerability assessments** with Risk and Compliance, applying **OWASP** findings and tracking remediation through release cycles.
- Wrote **PowerShell**, **Bash**, and **Python** scripts for certificate renewal, scheduled maintenance, and log export, reducing repetitive operational work for the platform team.

DevOps Engineer CO-OP, On-Site, **Midh Technologies**, TX

June 2023 - Dec 2023

- Managed production infrastructure for ML workloads, supporting **GPU backed Kubernetes clusters** on **AWS EKS** with cost optimization strategies including node pool configuration, resource quotas, and auto scaling policies.
- Designed high-performance inference infrastructure deploying **FastAPI** endpoints behind **AWS API Gateway** with **IAM authentication**, request throttling, and **Lambda** integrations for scalable ML inference.
- Developed infrastructure automation through **Ansible playbook** development, reducing manual server configuration and deployment time by **50%** while ensuring configuration consistency.
- Built production **CI/CD pipelines** with **GitHub Actions**, automating testing, Docker image builds, security scanning, and deployments to **AWS Amplify** and **EKS** environments.
- Implemented infrastructure security practices including **RBAC** policies, **IAM role** management, and **VPC** security configurations across AWS services and application layers.
- Provisioned **AWS** and **Azure** environments with **Terraform**, including **EKS/AKS** clusters, **API Gateway/Azure API Management**, **S3**, and **PostgreSQL**, enabling repeatable **environment promotion**.
- Extended pipelines to **Azure DevOps** for **Azure App Service** deployments of the **Next.js** company platform, adding **pytest** gates and rollback steps.
- Operated scheduled **Python ETL** jobs loading **PostgreSQL** and **MongoDB**, with containerized runs, retries, and **CloudWatch** alerting on failures.

DevOps/Platform Engineer, On-Site, **Compendious Medialabs Pvt. Ltd.**, IN

June 2021 - July 2022

- Managed hybrid infrastructure for **on premise Linux servers** (Ubuntu) with containerized deployments using **Docker**, establishing infrastructure reliability and automation through **Bash** scripting and configuration management.
- Designed and implemented self hosted CI/CD infrastructure deploying **GitHub Actions runners** on local infrastructure, building production grade pipelines for automated testing, Docker builds, and multiple environment deployments.
- Established production grade reverse proxy infrastructure with **Nginx** and **SSL/TLS** termination, optimizing external service exposure with **250ms** average response time and high availability.

- Implemented observability strategy deploying **Prometheus** and **Grafana** monitoring on local infrastructure; automated infrastructure tasks with **Python/Bash** scripting, reducing manual operational overhead by **60%**.
- Added **Jenkins** pipelines alongside **GitHub Actions** for legacy services, standardizing **Git branching strategies**, build artifacts, and release tagging.
- Containerized **FastAPI** and **React** applications and scheduled **Python** scraping/ETL jobs (**Selenium**, **Pandas**) with **Docker** and **Cron**, with logging and failure alerts.
- Hardened **Linux** hosts with firewall rules, **SSH** key access, patching schedules, and automated backups, and kept **WebSocket** streaming services stable behind **Nginx**.

Software Engineer / DevOps, Remote, VRoKCs, MO

March 2020 - May 2021

- Managed production infrastructure on **Azure**, deploying **Flask web applications** on **Azure App Services** with high availability configurations, auto scaling, and integrated monitoring.
- Built infrastructure automation through **Azure DevOps** pipelines for ML model deployment, implementing artifact storage with **Azure Blob Storage** and version control workflows.
- Designed containerized infrastructure using **Azure Container Instances** and **Docker** for ML inference services, ensuring environment parity between development and production with consistent deployment practices.
- Developed CI/CD infrastructure using **Azure DevOps** for ML lifecycle management, automating training and deployment workflows and reducing deployment cycles by **40%** through pipeline optimization.
- Deployed **Flask** and **TensorFlow Lite** inference services with **Docker**, keeping image upload, webcam streaming, and live prediction endpoints responsive under load.
- Configured **Azure Monitor** alerts, App Service scaling rules, and deployment slots for zero-downtime swaps across compute, storage, and networking.
- Added **automated testing** and **secure coding** checks to release pipelines for critical-path **Flask** APIs, and partnered with ML engineers on **Python** preprocessing pipelines for data collection and dataset preparation.

Software Developer / Build & Release, On-Site, Shree Hari Marketing PVT. LTD., IN

March 2019 - Feb 2020

- Set up **Jenkins** build jobs and **Git** versioning with release checklists for **environment promotion** across Dev, Test, and Production for order, inventory, and customer services.
- Deployed and maintained **Python Flask** and **Node.js REST APIs** and the **Angular** UI on **Linux** servers behind **Nginx**, with service restarts, log rotation, and health checks.
- Automated **SQL Server** backups, reporting, and data cleanup with **Python** and **Bash** scripts scheduled through **Cron**, reducing manual operational tasks.
- Ran unit and integration **automated testing**, **code reviews**, and **secure coding** checks (input validation, auth, logging) as release gates before production deployments.

Web/Game Developer, On-Site, underDogs Gaming Studios, IN

Jan 2018 - Feb 2019

- Contributed to backend API infrastructure for gaming services, integrating and optimizing **Quarters API** across multiple games with secure transaction, supporting **load balancing** and **API rate limiting** for high availability systems.
- Supported infrastructure reliability initiatives by optimizing real time client server communication, validating and refining backend interaction flows using **Python** and **C#**, implementing error handling and health monitoring.
- Developed unified transaction infrastructure connecting external services (**payments**, **inventory**, **shipping**) to internal game systems, following consistent API patterns and infrastructure best practices.
- Participated in infrastructure security and testing practices, implementing **API security best practices**, comprehensive logging, and integration testing frameworks for edge cases including payment failures and network resilience.
- Stood up release and hosting for game backends with **Git** source control, scripted deployments, and **health checks** for **Python** and **Node.js** services.

Education

State University of New York at Binghamton

Master of Science in Computer Science

Aug 2022 - Dec 2023

3.44/4.00 GPA

Certifications



Microsoft Certified: DevOps Engineer Expert

[View Credential](#)

AZ-400: Designing and Implementing Microsoft DevOps Solutions
AZ-900: Microsoft Certified: Azure Fundamentals
Google IT Automation with Python (Specialization)

[View Credential](#)